F12 4th CAR

Lecture 36: Section 5.5
Multiple Angles and Product-to-Sum Formulas

Double-angle formulas

Power reducing formulas

Half-angle formulas

Memorize

Product-to-sum formulas

Sum-to-product formulas

For this class you are not required to memorize these.

Double-Angle Formulas Memorize

1. $\sin 2u = 2\sin u \cos u$

2. $\cos 2u = \overbrace{\cos^2 u}^{1-\sin^2 u} - \sin^2 u$ $\quad 1-\cos^2 u$

$$= 1 - 2\sin^2 u$$

$$= 2\cos^2 u - 1$$

3. $\tan 2u = \dfrac{2\tan u}{1-\tan^2 u} = \dfrac{\sin 2u}{\cos 2u}$

Proof.

(1) $\boxed{\sin 2u} = \sin(u+u) = \sin u \cos u + \cos u \sin u$

$= \boxed{2\sin u \cos u}$

(2) $\boxed{\cos 2u} = \cos(u+u) = \cos u \cos u - \sin u \sin u$

$= \boxed{\cos^2 u - \sin^2 u}$

$\sin 2u = 2\sin u \cos u$

NOTE: The double-angle formulas are not restricted to angles u and $2u$. For example:

$\sin 6\theta = \sin(2(3\theta)) = \boxed{2\sin 3\theta \cos 3\theta}$

$\cos 4\theta = \cos(2(2\theta)) = \cos^2 2\theta - \sin^2 2\theta$

or $= 1 - 2\sin^2 2\theta$

or $= 2\cos^2 2\theta - 1$

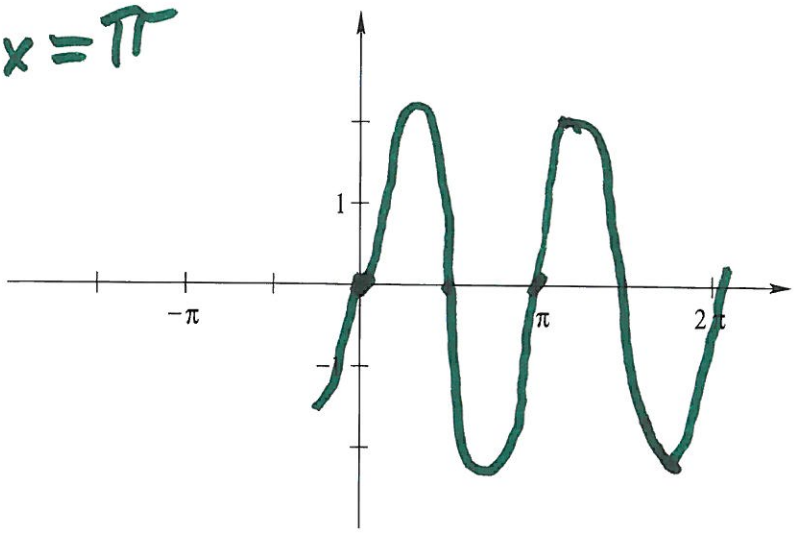
ex. Graph $y = 4\sin x \cos x$.

$y = 2 \cdot \underbrace{2\sin x \cos x}$

$y = 2\sin 2x$

$2x = 0 \rightarrow x = 0$

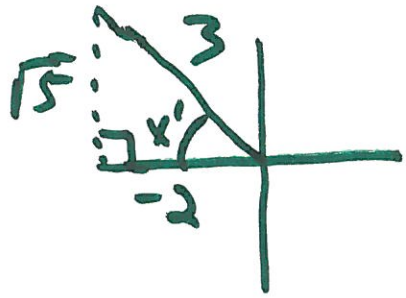
$2x = 2\pi \rightarrow x = \pi$



ex. If $\cos x = -\frac{2}{3}$ and x is in quadrant II, find

1) $\sin 2x = 2\sin x\cos x$

formula



$= 2\left(\frac{\sqrt{5}}{3}\right)\left(-\frac{2}{3}\right)$

$= \boxed{\frac{-4\sqrt{5}}{9}}$

formula

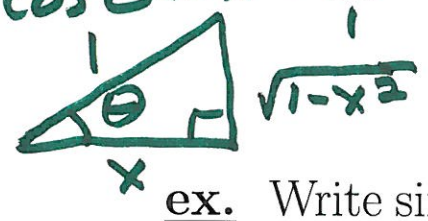
2) $\tan 2x = \frac{2\tan x}{1-\tan^2 x} = \frac{2\left(-\frac{\sqrt{5}}{2}\right)}{1-\left(-\frac{\sqrt{5}}{2}\right)^2}$

$= \frac{-\sqrt{5}}{1-\frac{5}{4}} = \frac{-\sqrt{5}}{-\frac{1}{4}} = \boxed{4\sqrt{5}}$

or $\frac{\sin 2x}{\cos 2x}$

① $\theta = \arccos x$

$\cos\theta = x = \frac{x}{1}$



② $\sin 2\theta = 2\sin\theta\cos\theta$

$= 2\left(\frac{\sqrt{1-x^2}}{1}\right)x$

ex. Write $\sin(2\overbrace{\arccos x}^{\theta})$ as an algebraic expression.

$= \boxed{2x\sqrt{1-x^2}}$

A Triple-Angle Formula

apply sum formula

ex. Write $\cos 3x$ in terms of $\cos x$.

formula

$\cos 3x = \cos(x+2x) = \cos x \cos 2x - \sin x \sin 2x$

$= \cos x(2\cos^2 x - 1) - \sin x(2\sin x\cos$

$= 2\cos^3 x - \cos x - 2\sin^2 x\cos x$

$= 2\cos^3 x - \cos x - 2(1-\cos^2 x)\cos x$

$= 2\cos^3 x - \cos x - 2(\cos x - \cos^3 x)$

$= 2\cos^3 x - \cos x - 2\cos x + 2\cos^3 x$

$= \boxed{4\cos^3 x - 3\cos x}$

Derivation of (1): Take cosine double angle formula and solve for $\sin^2 u$.

$\cos 2u = 1 - 2\sin^2 u \rightarrow 2\sin^2 u = 1 - \cos 2u$

$\sin^2 u = \dfrac{1-\cos 2u}{2}$

Power-Reducing Formulas

1. $\sin^2 u = \dfrac{1 - \cos 2u}{2}$

2. $\cos^2 u = \dfrac{1 + \cos 2u}{2}$

3. $\tan^2 u = \dfrac{1 - \cos 2u}{1 + \cos 2u}$

Use $\cos 2u = 2\cos^2 u - 1$ and solve for $\cos^2 u$.

$2\cos^2 u = \cos 2u + 1$

$\cos^2 u = \dfrac{\cos 2u + 1}{2}$

ex. Express $\sin^2 x \cos^2 x$ in terms of the first power of cosine.

$\sin^2 x \cdot \cos^2 x = \left(\dfrac{1-\cos 2x}{2}\right) \cdot \left(\dfrac{1+\cos 2x}{2}\right) =$ FOIL

$= \dfrac{1 - \cos^2 2x}{4} = \dfrac{1}{4}\left[1 - \cos^2 2x\right] =$

$= \dfrac{1}{4}\left[1 - \dfrac{1+\cos 4x}{2}\right] = \dfrac{1}{4}\left[1 - \left(\dfrac{1}{2} + \dfrac{\cos 4x}{2}\right)\right] =$

$= \dfrac{1}{4}\left[1 - \dfrac{1}{2} - \dfrac{\cos 4x}{2}\right] = \dfrac{1}{4}\left(\dfrac{1}{2} - \dfrac{\cos 4x}{2}\right) = \boxed{\dfrac{1}{8} - \dfrac{\cos 4x}{8}}$

Memorize

Half-Angle Formulas Memorize

1. $\sin\frac{u}{2} = \pm\sqrt{\frac{1-\cos u}{2}}$

2. $\cos\frac{u}{2} = \pm\sqrt{\frac{1+\cos u}{2}}$

3. $\tan\frac{u}{2} = \frac{1-\cos u}{\sin u} = \frac{\sin u}{1+\cos u}$

The choice of the + or − sign depends on the quadrant in which $u/2$ lies.

QI so (+)

ex. Find the exact value of $\sin 22.5°$. note: $22.5 = \frac{45}{2}$

$\sin 22.5° =$

$= \sin\frac{45}{2} = \sqrt{\frac{1-\cos 45°}{2}} =$

$= \sqrt{\frac{\left(1-\frac{\sqrt{2}}{2}\right)\left(\frac{2}{2}\right)}{2}} = \sqrt{\frac{2-\sqrt{2}}{4}} = \boxed{\frac{\sqrt{2-\sqrt{2}}}{2}}$

Q II so (−)

ex. Find the exact value of $\cos 165°$. $= \cos\left(\frac{330}{2}\right)^\circ$

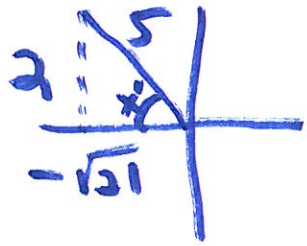
$$\cos 165^\circ = \cos\left(\frac{330}{2}\right)^\circ = -\sqrt{\frac{1+\cos 330^\circ}{2}}$$

$$= -\sqrt{\left(\frac{1+\frac{\sqrt{3}}{2}}{2}\right)\frac{2}{2}} = -\sqrt{\frac{2+\sqrt{3}}{4}} = \boxed{\frac{-\sqrt{2+\sqrt{3}}}{2}}$$

ex. Find $\tan\frac{x}{2}$ if $\sin x = \frac{2}{5}$ and x is in quadrant II.

formula #3

$$\tan\frac{x}{2} = \frac{1-\cos x}{\sin x} =$$



$$= \frac{1-\left(\frac{-\sqrt{21}}{5}\right)}{\frac{2}{5}} = \cdots = \boxed{\frac{5+\sqrt{21}}{2}}$$

No time to cover in lecture - put on Sakar pages 9-13

Don't need to memorize

Product-to-Sum Formulas

$$\sin u \sin v = \frac{1}{2}[\cos(u-v) - \cos(u+v)]$$

$$\cos u \cos v = \frac{1}{2}[\cos(u-v) + \cos(u+v)]$$

$$\sin u \cos v = \frac{1}{2}[\sin(u+v) + \sin(u-v)]$$

$$\cos u \sin v = \frac{1}{2}[\sin(u+v) - \sin(u-v)]$$

Don't need to memorize

Sum-to-Product Formulas

$$\sin u + \sin v = 2\sin\left(\frac{u+v}{2}\right)\cos\left(\frac{u-v}{2}\right)$$

$$\sin u - \sin v = 2\cos\left(\frac{u+v}{2}\right)\sin\left(\frac{u-v}{2}\right)$$

$$\cos u + \cos v = 2\cos\left(\frac{u+v}{2}\right)\cos\left(\frac{u-v}{2}\right)$$

$$\cos u - \cos v = -2\sin\left(\frac{u+v}{2}\right)\sin\left(\frac{u-v}{2}\right)$$

sum-to-product since $\frac{75+15}{2} = \frac{90}{2} = 45$

$\frac{75-15}{2} = \frac{60}{2} = 30$

ex. Find the exact value:

1) $\sin 75° + \sin 15°$

sum to product formula

$\sin u + \sin v = 2\sin\left(\frac{u+v}{2}\right)\cos\left(\frac{u-v}{2}\right)$

$\sin 75° + \sin 15° = 2\sin\left(\frac{75+15}{2}\right)\cos\left(\frac{75-15}{2}\right) =$

$= 2\sin 45° \cos 30°$

$= 2\,\frac{\sqrt{2}}{2}\cdot\frac{\sqrt{3}}{2} = \boxed{\frac{\sqrt{6}}{2}}$

sum-to-product since $\frac{\frac{\pi}{12}+\frac{5\pi}{12}}{2} = \frac{\pi}{4}$ and $\frac{\frac{\pi}{12}-\frac{5\pi}{12}}{2} = -\frac{\pi}{6}$

2) $\cos\frac{\pi}{12} + \cos\frac{5\pi}{12}$ $= \frac{\sqrt{6}}{2}$

sum to product formula

Don't work in class.

$\cos u + \cos v = 2\cos\left(\frac{u+v}{2}\right)\cos\left(\frac{u-v}{2}\right)$

$\cos\frac{\pi}{12} + \cos\frac{5\pi}{12} = 2\cos\left(\frac{\frac{\pi}{12}+\frac{5\pi}{12}}{2}\right)\cos\left(\frac{\frac{\pi}{12}-\frac{5\pi}{12}}{2}\right)$

$= 2\cos\frac{\pi}{4}\cos\left(-\frac{\pi}{6}\right)$

$= 2\left(\frac{\sqrt{2}}{2}\right)\left(\frac{\sqrt{3}}{2}\right) = \boxed{\frac{\sqrt{6}}{2}}$

Product to sum since $37.5 + 7.5 = 45°$

$37.5 - 7.5 = 30°$

3) $\cos 37.5° \sin 7.5° = \frac{\sqrt{2}-1}{4}$

$\cos u \sin v = \frac{1}{2}[\sin(u+v) - \sin(u-v)]$

$\cos 37.5 \sin 7.5 = \frac{1}{2}[\sin(37.5+7.5) - \sin(37.5-7.5)]$

$= \frac{1}{2}[\sin 45° - \sin 30°]$

$= \frac{1}{2}\left(\frac{\sqrt{2}}{2} - \frac{1}{2}\right) = \frac{1}{2}\left(\frac{\sqrt{2}-1}{2}\right)$

$= \boxed{\frac{\sqrt{2}-1}{4}}$

ex. Verify: $\dfrac{\sin 3x - \sin x}{\cos 3x + \cos x} = \tan x$

Sum-to-product formulas

$\sin u - \sin v = 2\cos\left(\frac{u+v}{2}\right)\sin\left(\frac{u-v}{2}\right)$

$\cos u + \cos v = 2\cos\left(\frac{u+v}{2}\right)\cos\left(\frac{u-v}{2}\right)$

$$LHS = \frac{\cancel{2}\cancel{\cos\left(\frac{3x+x}{2}\right)} \cdot \sin\left(\frac{3x-x}{2}\right)}{\cancel{2}\cancel{\cos\left(\frac{3x+x}{2}\right)} \cdot \cos\left(\frac{3x-x}{2}\right)} = \frac{\sin x}{\cos x}$$

$= \tan x = RHS$

Solving Trigonometric Equations

ex. Solve the equation in the interval $[0, 2\pi)$.

Need same angle

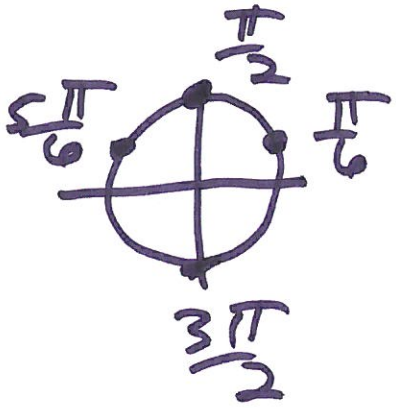
1) $\sin 2x - \cos x = 0$

$2\sin x\cos x - \cos x = 0$

$\cos x(2\sin x - 1) = 0$

$\cos x = 0 \qquad 2\sin x - 1 = 0$

$\sin x = \frac{1}{2}$



$\boxed{x = \frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6}, \frac{3\pi}{2}}$

Check solution since possible domain problem.

Need same angle.

Half-angle formula

2) $\tan\frac{x}{2} - \sin x = 0$

$\sin x\left[\frac{1-\cos x}{\sin x} - \sin x = 0\right]$

$1 - \cos x - \sin^2 x = 0$

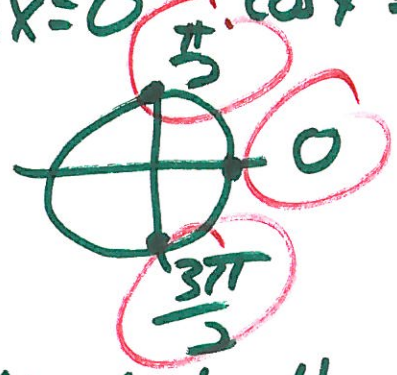
$1 - \cos x - (1 - \cos^2 x) = 0$

$1 - \cos x - 1 + \cos^2 x = 0$

$\cos^2 x - \cos x = 0$

$\cos x(\cos x - 1) = 0$

$\cos x = 0 \qquad \cos x =$



Need to check all!

$x = 0$

$\tan\frac{0}{2} - \sin 0 = 0$

$\tan 0 - \sin 0 = 0$

$0 - 0 = 0$ ✓

They all check

sum-to-product formula

$\cos u + \cos v = 2\cos\left(\frac{u+v}{2}\right)\cos\left(\frac{u-v}{2}\right)$

3) $\cos 4x + \cos 2x = \cos x$

$LHS = \cos 4x + \cos 2x = 2\cos\left(\frac{4x+2x}{2}\right)\cos\left(\frac{4x-2x}{2}\right)$

$= 2\cos 3x \cos x$

thus $\underbrace{\cos 4x + 2x}_{2\cos 3x \cos x} = \cos x$

$2\cos 3x \cos x = \cos x$

$2\cos 3x \cos x - \cos x = 0$

$\cos x(2\cos 3x - 1) = 0$

$\cos x = 0$

$\pi/2$

$\frac{3\pi}{2}$

$2\cos 3x - 1 = 0$

$\cos 3x = \frac{1}{2}$

$\frac{\pi}{3}$

$\frac{5\pi}{3}$

$3x = \frac{\pi}{3} + 2n\pi$

$x = \frac{\pi}{9} + \frac{2n\pi}{3}$

$3x = \frac{5\pi}{3} + 2n\pi$

$x = \frac{5\pi}{9} + \frac{2n\pi}{3}$

over

$$x = \frac{\pi}{9} + \frac{2\pi n}{3}$$

n	0	1	2	3
x	$\frac{\pi}{9}$	$\frac{7\pi}{9}$	$\frac{13\pi}{9}$	$\cancel{\frac{19\pi}{9}}$ +∞ b.g

$$x = \frac{5\pi}{9} + \frac{2n\pi}{3}$$

x	0	1	2	3
y	$\frac{5\pi}{9}$	$\frac{11\pi}{9}$	$\frac{17\pi}{9}$	$\cancel{\frac{23\pi}{9}}$ +∞ b.g

$$\boxed{x = \frac{\pi}{9}, \frac{5\pi}{9}, \frac{7\pi}{9}, \frac{11\pi}{9}, \frac{13\pi}{9}, \frac{17\pi}{9}, \frac{3\pi}{2}}$$